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ME 5130: Advanced Heat and Mass Transfer
Homework 01: Heat Conduction
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1–2, Problem Statement: Verify that ∇T and $\nabla \cdot \mathbf{q}$ in the spherical coordinate system (r, ϕ, θ) are given as:

$$\nabla T = \hat{\mathbf{u}}_{\mathbf{r}} \frac{\partial T}{\partial r} + \hat{\mathbf{u}}_{\phi} \frac{1}{r \sin\theta} \frac{\partial T}{\partial \phi} + \hat{\mathbf{u}}_{\theta} \frac{1}{r} \frac{\partial T}{\partial \theta} \tag{1}$$

$$\nabla \cdot \mathbf{q} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 q_r \right) + \frac{1}{r \sin\theta} \frac{\partial q_\phi}{\partial \phi} + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} \left(q_\theta \sin\theta \right) \tag{2}$$

1–2, Solution:

1–12, Problem Statement: Set up the mathematical formulation of the following head conduction problems:

1. A slab in $0 \le x \le L$ is initially at a temperature $F(x)$. For times $t > 0$, the boundary at $x = 0$ is kept insulated and the boundary at $x = L$ dissipates heat by convection into a medium at zero temperature.

2. A semiinfinite region $0 \le x < \infty$ is initially at a temperature $F(x)$. For times $t > 0$, heat is generated in the medium at a constant rate of $g_0 W/m^3$, while the boundary at x = 0 is kept at zero temperature.

3. A solid cylinder $0 \le r \le b$ is initially at a temperature $F(r)$. For times $t > 0$, heat is generated in the medium at a rate of $g(r)$, W/m^3, while the boundary at $r = b$ dissipates heat by convection into a medium at zero temperature.

4. A solid sphere $0 \le r \le b$ is initially at temperature $F(r)$. For times $t > 0$, heat is generated in the medoum at a rate of $g(r)$, W/m^3, while the boundary at $r = b$ is kept at a uniform temperature T_0.

1–12, Solution:

1–15, Problem Statement: A long cylindrical iron bar of diameter $D = 5\,\mathrm{cm}$, initially at temperature $T_0 = 650\,°\mathrm{C}$, is exposed to an air stream at $T_\infty = 50\,°\mathrm{C}$. The heat transfer coefficient between the air stream and the surface of the bar is $h = 80\,\mathrm{W\,m^{-2}\,°C^{-1}}$. Thermophysical properties may be taken as $\rho = 7800\,\mathrm{kg\,m^{-3}}$, $C_p = 460\,\mathrm{J\,kg^{-1}\,°C^{-1}}$, and $k = 60\,\mathrm{W\,m^{-1}\,°C^{-1}}$. Determine the time required for the temperature of the bar to reach $250\,°\mathrm{C}$ by using the lumped system analysis.

1–15, Solution:

1–16, Problem Statement: A thermocouple is to be used to measure the temperature in a gas stream. The junction may be approximated as a sphere having thermal conductivity $k = 25\,\mathrm{W\,m^{-1}\,°C^{-1}}$, $\rho = 8400\,\mathrm{kg\,m^{-3}}$, and $C_p = 0.4\,\mathrm{kJ\,kg^{-1}\,°C^{-1}}$. The heat transfer coefficient between the junction and the gas stream is $h = 560\,\mathrm{W\,m^{-2}\,°C^{-1}}$. Calculate the diameter of the junction if the thermocouple should measure 95% of the applied temperature difference in 3s.

1–16, Solution:

2–2, Problem Statement: A slab, $0 \leq x \leq L$, is initially at a temperature $F(x)$. For times $t > 0$ the boundary surface at $x = 0$ is kept insulated and that at $x = L$ dissipates heat by convection into a medium at zero temperature with a heat transfer coefficient h. Obtain an expression for the temperature distribution $T(x, t)$ in the slab for times $t > 0$ and for the heat flux at the boundary surface $x = L$.

2–2, Solution:

2–6, Problem Statement: In a one-dimensional infinite medium, $-\infty < x < \infty$, initially, the region $a < x < b$ is at a constant temperature T_0, and everywhere outside this region is at zero temperature. Obtain an expression for the temperature distribution $T(x, t)$ in the medium for times $t > 0$.

2–6, Solution:

2–15, Problem Statement: Obtain an expression for the steady-state temperature distribution $T(x, y)$ in a rectangular region $0 \leq x \leq a$, $0 \leq y \leq b$ for the following boundary conditions: the boundary at $x = 0$ is kept insulated, the boundary at $y = 0$ is kept at temperature $f(x)$ and the boundaries at $x = a$ and $y = b$ dissipate heat by convection into an environment at zero temperature. Assume the heat transfer coefficient to be the same for both boundaries.

2–15, Solution: